THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 4: **Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student brings the north pole of a bar magnet near the north end of a compass needle. The needle's north end will: 1

(a) stay unchanged **(b)** move closer **(c)** turn away **(d)** melt

2. Which action would not demagnetise a bar magnet? 1

(a) heating it red hot **(b)** hammering it

(c) keeping it with a keeper **(d)** dropping repeatedly

3. The strength of a compass reading is best when it is placed: 1

(a) near speakers **(b)** near a fridge

(c) away from iron and magnets **(d)** near a steel desk

4. Which one of the following sets lists items that all can be stuck to a fridge magnet? 1

(a) plastic cap, wooden peg **(b)** copper coin, brass key

(c) glass bead, rubber **(d)** steel clip, iron nail

5. If the poles of a magnet are painted, how would you check the marking? 1

(a) suspend it and see which end points north

(b) taste it

(c) weigh it

(d) look at colour

6. Which of the following devices works because of magnets? 1

(a) a plastic pen **(b)** a loudspeaker **(c)** a candle **(d)** a spoon

7. A magnet is cut into three equal pieces. The number of magnets obtained is: 1

(a) 2 **(b)** 1 **(c)** 3 **(d)** 6

8. A bar attracts the north end of a compass needle at one end and the south end of the needle at the other end. The bar is most likely: 1

(a) unmagnetised iron **(b)** wood

(c) a magnet **(d)** plastic

9. **Assertion (A):** The attraction of a magnet is strongest at its poles. 1

Reason (R): Iron filings stick mostly near the ends of a bar magnet.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** A magnet always has two poles even if it is cut into two pieces. 1
Reason (R): Magnetic poles can be separated to get a single north pole.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Give reasons why a magnet is kept on the tabletop away from iron objects when not used. 2

12. Explain why magnetic poles always occur in pairs. 2

13. Why is a ceramic or plastic container used for storing magnets in a lab? 2

14. Why can a magnet pick a pin from a pile faster than hands? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If you drop a magnet onto the floor repeatedly, what will happen and why? 3

16. How are magnets useful in the kitchen? 3

17. How can you demonstrate that magnets lose power on heating? 3

OR

How will you know if a cord of iron wire is magnetised or not? 3

18. What happens when a very strong magnet is brought near a compass? 3

OR

Why should a suspended magnet be kept away from a steel table when finding directions? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Two bar magnets were placed on a table, with the ends marked N and S. When Dev brought the N end of one magnet near the N end of the other, they pushed apart. When he turned one magnet around, they pulled together and stuck. 5
(i) What happens when like poles are brought near each other?
(ii) What happens with unlike poles?
(iii) Name the ends of a magnet where attraction is strongest.
(iv) A magnet is kept with its S end near a suspended magnet's S end. Predict what happens and give a reason.

20. Explain how the compass was an important invention and how its working relates to the Earth's magnetism. 5

OR

Explain the following with examples: temporary magnet, permanent magnet, natural magnet. 5

END OF QUESTION PAPER — Total: 40 Marks